TEAS 7 Science Bootcamp

Anatomy & Physiology + Nursing Science Review — 2 Days × 4 Hours

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The exam at a glance

Science = 50 questions, 60 minutes (~1 min each). A&P is 18 of the 44 scored — so A&P first.

Topic	Questions	Priority
Anatomy & Physiology	18	★★★
Biology	9	★★
Scientific Reasoning	9	★★
Chemistry	8	★

- **Target:** 75%+ in science. No partial credit on "select all" items.
- **Expect:** hot-spot (click the organ), ordered-response (drag blood flow into order), fill-in.
- **Study loop:** read → cover → recall → practice questions → revisit in 2 days.

Bootcamp agenda: 2 days × 4 hours

Each hour = **45 min teach + 10 min quick quiz (or game realm) + 5 min break.** Lunch and breaks between days are outside the 8 hours.

Day 1 — Foundations and the "big four" systems

Hour	Topic	Students leave able to…
1	Exam at a glance + A&P foundations	Name planes, directions, tissues, feedback types
2	Heart & blood + lungs	Recite blood flow through the heart; explain acidosis vs alkalosis
3	Nervous + endocrine	Compare sympathetic vs parasympathetic; match glands to hormones
4	Digestive + urinary	Trace food and urine; explain ADH and aldosterone

Day 2 — Remaining systems, biology, chemistry and reasoning

Hour	Topic	Students leave able to…
5	Bones, muscles, skin	Explain blast vs clast, muscle types, skin layers and glands
6	Immune/lymphatic + reproductive	Fill in the active/passive immunity grid; name antibody types
7	Biology + chemistry	Match organelles; do a Punnett square; read pH; name reaction types

Hour	Topic	Students leave able to…
8	Scientific reasoning (15 min) + timed practice quiz (30 min) + cram-sheet review (15 min)	Identify variables; score 75%+ on the quiz

Start of Day 2: spend the first 5 minutes of Hour 5 on a rapid recall of Day 1 (blood flow, gland–hormone pairs, food and urine paths).

The bootcamp game: Realms of Anatomica

Each bootcamp hour has a matching game realm with **50 questions (400 in total)**. Students play the realm during that hour's quiz time.

Click: *Realms of Anatomica*

- **Day 1:** Realms 1–4 (Gloomhallow Manor, Emberheart Caldera, Stormspire Nexus, The Gumdrop Gut Works).
- **Day 2:** Realms 5–8 (The Rattling Gymnasium, Bastion of Wards, The Mad Alchemist's Lab, The Sandglass Archive).
- **Boss battle:** 6 correct answers defeat the boss; 3 wrong answers (or timeouts) end the attempt. Questions are drawn at random from the realm's 50, so every attempt is different.
- **Scoring:** each attempt is scored as % correct. All attempts are averaged, and students choose which attempt counts as **final**.
- **3 attempts per realm.** The next realm opens only after a final attempt is locked in. Day 2 opens after Hour 4.
- **Save & continue:** students can pause any battle with **Save & exit** and resume within **24 hours** with their points and answers kept. After 24 hours the attempt closes with the score reached.
- **Mystery chests** appear once per battle and give a bonus (extra heart, extra time, points, or two wrong answers removed).
- **Report:** after all 8 realms, students download a **PDF report** (attempt scores, averages, finals and topics to review) and submit it.

- **No student sign-in.** Students type their full name, student ID or school email; if they are on an uploaded class list, the game recognizes them and adds their school, program and cohort to the report.
- **Instructor login** (landing page): instructors sign in with their email and password to upload class lists (CSV), manage several schools, programs and cohorts, reset a student, a group or everyone, and add or revoke instructors. Main admins: jphernandezrn@gmail.com and stellarlifetech@gmail.com.
- Students should play both days on the **same device and browser**, because progress is saved there.

A&P foundations (Day 1 · Hour 1)

Always picture anatomical position: standing, facing you, palms forward. Left/right = the *patient's*.

Direction pairs

- **Superior / inferior** = head / feet
- **Anterior / posterior** = front / back
- **Medial / lateral** = toward / away from midline
- **Proximal / distal** = near / far from where a limb attaches *(limbs only)*
- **Superficial / deep** = surface / inside

Planes — "Split Front Top"

- **S**agittal → left | right
- **F**rontal (coronal) → front | back
- **T**ransverse → top | bottom

Cavities: Dorsal = brain + spinal cord. Ventral = thoracic (heart, lungs) + abdominopelvic, split by the **diaphragm**. Kidneys sit *behind* the peritoneum (retroperitoneal).

Quadrants: RUQ liver/gallbladder · LUQ stomach/spleen · RLQ appendix · LLQ sigmoid colon.

Levels: "**C**ats **C**an **T**ake **O**ver **O**ur **O**ffice" = Chemical → Cell → Tissue → Organ → Organ system → Organism.

4 tissues — "ECMN" ("Every Cell Must Network")

- **E**pithelial — covers and lines (skin, gut lining); no blood vessels
- **C**onnective — supports (bone, blood, fat, tendons)
- **M**uscle — moves
- **N**ervous — signals

Feedback

- **Negative** = reverses the change (most common): hot → sweat; high sugar → insulin.
- **Positive** = amplifies until done: **labor (oxytocin), clotting, breastfeeding**. "Positive = **B**abies and **B**lood."

Body systems (Day 1 · Hours 2–4 and Day 2 · Hours 5–6)

Each system: **what it does**, **the must-knows**, **a memory hook**, and a **Nursing link**.

Heart & blood (cardiovascular)

Blood flow — memorize this order:

Body → vena cava → **R atrium** → tricuspid → **R ventricle** → pulmonary valve → pulmonary artery → **LUNGS** → pulmonary vein → **L atrium** → mitral → **L ventricle** → aortic valve → aorta → body

- Valves in order: "**T**ry **P**ulling **M**y **A**orta" (Tricuspid, Pulmonary, Mitral, Aortic).
- **A**rteries go **A**way from the heart. Veins return.
- **Trick question:** pulmonary *artery* = deoxygenated; pulmonary *vein* = oxygenated.
- **Left ventricle** = thickest wall (pumps to the whole body).
- **SA node** = pacemaker. Capillaries = exchange (1 cell thick).
- RBC = O_2 (hemoglobin, iron) · WBC = infection · platelets = clotting · plasma = 55%, mostly water.
- BP normal < 120/80. Systolic = squeeze, diastolic = relax.

Nursing link: *Left* heart failure → fluid in **L**ungs. *Right* heart failure → swollen legs.

Lungs (respiratory)

Air path: nose → pharynx → larynx → trachea → bronchi → bronchioles → **alveoli** (gas exchange by diffusion).

- "**P**lease **L**et **T**he **B**reath **B**e **A**ir" = Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveoli.
- **Diaphragm** contracts *down* → air rushes *in*.
- Breathing is driven by **CO_2** (sensed by the **medulla**).
- Epiglottis = lid that keeps food out of the airway.
- Right lung 3 lobes, left lung 2 (room for the heart).
- **Slow breathing** keeps CO_2 → acidosis. **Fast breathing** blows off CO_2 → alkalosis.

Nursing link: Normal blood pH = **7.35–7.45**. Below = acidosis, above = alkalosis.

Nervous system

- **CNS** = brain + spinal cord. **PNS** = all other nerves.
- **Sympathetic** = **fight or flight** (↑ HR, ↑ BP, big pupils, digestion stops).
- **Parasympathetic** = **rest and digest** (↓ HR, digestion on; vagus nerve).
- **Neuron: D**endrite → **C**ell body → **A**xon ("**D**on't **C**all **A**gain" — signal flows one way). Myelin speeds it up.
- **SAME: S**ensory = **A**fferent (to the brain), **M**otor = **E**fferent (away).
- Cerebrum = thinking · **cerebellum = balance** · **medulla = breathing/HR** · hypothalamus = temperature, thirst, controls pituitary.

Nursing link: Left-brain stroke → right-side weakness (brain controls the opposite side).

Endocrine (hormones)

Gland	Remember
Hypothalamus	**Makes** ADH + oxytocin; bosses the pituitary
Anterior pituitary	"Master gland": GH, TSH, ACTH, FSH, LH, prolactin
Posterior pituitary	**Stores/releases** ADH + oxytocin
Thyroid	T3/T4 = metabolism. Calci**tonin tones down** calcium
Parathyroid	PTH **pulls** calcium **up** from bone
Adrenal cortex	Cortisol (stress, sugar), aldosterone (keep salt)
Adrenal medulla	Epinephrine = adrenaline
Pancreas	**Insulin** lowers sugar (lets it *in* to cells); **glucagon** raises it ("glucose **gone**" → make more)

- Steroid hormones (cortisol, estrogen, testosterone) are made from **cholesterol**.
- Most hormones are controlled by **negative feedback**.

Nursing link: Type 1 diabetes = no insulin. Type 2 = insulin doesn't work well.

Digestive system

Food path: mouth → esophagus → stomach → small intestine (**D**uodenum, **J**ejunum, **I**leum = "**D**ow **J**ones **I**ndustrial") → large intestine → rectum.

- Mouth: **amylase** starts **carbs**.
- Stomach: acid + **pepsin** starts **protein**.
- **Liver makes bile, gallbladder stores it.** Bile **emulsifies fat** (not an enzyme).
- Pancreas sends enzymes for all three (amylase, lipase, trypsin).
- **Small intestine absorbs nutrients** (villi). **Large intestine absorbs water.**

Nursing link: RUQ pain after fatty food → think gallbladder. RLQ pain → think appendix.

Urinary system

Urine path: kidney → ureter → bladder → urethra ("**K**ids **U**se **B**ig **U**mbrellas").

- **Nephron** = filtering unit. Steps: **F**ilter (glomerulus) → **R**eabsorb → **S**ecrete → **E**xcrete ("**F**ilter, **R**eclaim, **S**pit, **E**xit").
- **ADH** = **A**nti-**D**iuretic = **holds water** → less urine. (Alcohol blocks ADH → you pee more.)
- **Aldosterone** = holds **salt** (and water follows) → ↑ BP.
- Kidneys make **EPO** (→ red blood cells) and **renin** (→ raises BP).

Nursing link: Glucose or protein in urine = abnormal. Kidney failure → high BP + anemia.

Skeletal system

- **206 bones.** Jobs: support, protect, move, store calcium, **make blood cells (red marrow)**.
- **Blast = Build. Clast = Crush.** (Osteoblast builds, osteoclast breaks down.)
- Bone types: **long** (femur) · **short** (wrist/ankle) · **flat** (skull, ribs, sternum) · **irregular** (vertebrae, pelvis) · **sesamoid** (kneecap).
- **L**igament = bone to bone ("**L**ink"). **T**endon = muscle to bone ("**T**ug").
- Vertebrae: "Breakfast at **7**, lunch at **12**, dinner at **5**" = cervical, thoracic, lumbar.

Nursing link: Osteoporosis = clasts beat blasts → fragile bones. Risk: menopause (low estrogen), low calcium/vitamin D, inactivity.

Muscular system

Type	Control	Striped?	Where
Skeletal	Voluntary	Yes	Bones
Cardiac	Involuntary	Yes	Heart only
Smooth	Involuntary	No	Organs, vessels

- **Contraction:** nerve signal → **calcium** released → **myosin** grabs **actin** and pulls → muscle shortens. Needs **ATP**.
- Muscles work in **pairs** (biceps bends, triceps straightens).
- **Ab**duction = **ab**ducted away from midline. **Add**uction = **add** back to the body.

Skin (integumentary)

- Layers top → down: **Epidermis** (no blood vessels; melanin, keratin) → **Dermis** (vessels, nerves, glands, hair) → **Hypodermis** (fat).
- Jobs: barrier, temperature control, sensation, **makes vitamin D**.
- Glands: **Eccrine** = everyday cooling sweat · **Apocrine** = armpit/groin, body odor (after puberty) · **Sebaceous** = **sebum** (oil).

Nursing link: Turn bedbound patients every 2 hours to prevent pressure injuries.

Immune & lymphatic

- **Lymphatic:** returns fluid to blood; lymph nodes filter; **spleen** filters blood; **thymus** trains **T** cells.
- **Innate** (born with, nonspecific): skin, mucus, stomach acid, fever, inflammation, phagocytes.
- **Adaptive** (learned, specific, has memory): **B cells → antibodies**. **T cells → kill infected cells**.

	You make the antibodies (lasts long)	You're *given* antibodies (short-term)
Natural	**Active:** you catch the disease	**Passive:** mom → baby (placenta, breast milk)
Artificial	**Active:** vaccine	**Passive:** antibody shot

Hook: Active = your body does the work. Passive = borrowed.

- Antibodies: **IgG** = most common, crosses placenta ("**G**estation") · **IgA** = tears, saliva, breast milk · **IgM** = first responder ("**M**ain first") · **IgE** = allergies ("**E**ek!").

Nursing link: HIV attacks helper T cells → frequent infections.

Reproductive system

- **Male:** testes make sperm + **testosterone** (the main androgen) → epididymis (matures) → vas deferens → urethra.
- **Female:** ovaries make eggs + estrogen/progesterone → **fertilization in the fallopian tube** → implants in the uterus.
- **LH surge → ovulation** (~day 14). Progesterone = "**pro-gestation**" (keeps the lining).
- Gametes are made by **meiosis** → 23 chromosomes each. Egg + sperm → 46.

Nursing link: Embryo implanted in the fallopian tube = **ectopic pregnancy** (emergency).

Biology (Day 2 · Hour 7)

Organelles — think of the cell as a factory

Organelle	Job	Factory role
Nucleus	Holds DNA	Boss's office
Mitochondria	Makes **ATP** (cellular respiration)	Power plant
Ribosomes	Make **proteins**	Workers
Rough ER	Protein transport (has ribosomes)	Assembly line
Smooth ER	Makes lipids, detox	Chemical shop
Golgi	Packages and ships proteins	Shipping/post office
Lysosome	Breaks down waste	Clean-up crew
Cell membrane	Controls what enters/leaves	Security gate
Cytoskeleton	Shape and movement (made of **protein**)	Building frame

Plant cells also have a **cell wall**, **chloroplasts** and a big central vacuole.

Mitosis vs meiosis

- **Mitosis** = **2** identical body cells, diploid (46). Growth and repair. Phases: **PMAT** (Prophase, Metaphase — line up in the **M**iddle, Anaphase — pull **A**part, Telophase — **T**wo nuclei).
- **Meiosis** = **4** unique sex cells, haploid (23). "Mei-**O**sis makes **O**va."

DNA → RNA → protein

- DNA pairs: **A–T, C–G** ("**A**pple in the **T**ree, **C**ar in the **G**arage"). RNA swaps T for **U** (A–U).
- **Transcription** = DNA → mRNA (in the nucleus). **Translation** = mRNA → protein (at the ribosome).
- DNA = double helix, deoxyribose. RNA = single strand, ribose.

Genetics

- Dominant = capital (**B**); recessive = lowercase (**b**). Recessive shows only as **bb**.
- Genotype = the letters. Phenotype = what you see.
- Bb × Bb → **1 BB : 2 Bb : 1 bb** → 3:1 dominant to recessive look (75% / 25%).
- **Codominance** = both show (AB blood). **Incomplete dominance** = blend (red × white = pink).

Macromolecules — "CLiP N"

Molecule	Building block	Job
Carbohydrates	Monosaccharides (glucose)	Quick energy
Lipids	Fatty acids + glycerol	Long-term energy, membranes, hormones
Proteins	**Amino acids**	Enzymes, structure, antibodies
Nucleic acids	Nucleotides	DNA/RNA (genetic info)

Microorganisms: bacteria (no nucleus; treated with antibiotics) · viruses (not cells; need a host; antibiotics don't work) · fungi (yeast, mold) · protozoa (malaria parasite).

Chemistry (Day 2 · Hour 7)

The atom — "PEN"

- **P**rotons = **P**ositive, in nucleus. **Atomic number** = # of protons (identifies the element).
- **E**lectrons = negative, orbit the nucleus. Neutral atom: electrons = protons.
- **N**eutrons = **N**eutral, in nucleus. **Mass number** = protons + neutrons.
- **Isotope** = same protons, different neutrons. **Ion** = gained/lost electrons.
- **Valence electrons** = outer shell; decide bonding.

Bonds

- **Ionic** = electrons **given** (metal + nonmetal, e.g. NaCl). "**I**onic = **I** give."
- **Covalent** = electrons **shared** (nonmetal + nonmetal, e.g. H_2O). "**Co** = cooperate."
- **Hydrogen bonds** = weak attractions between water molecules (high surface tension and boiling point).
- **Electronegativity** rises **up and to the right** on the periodic table (fluorine highest).

Reaction types

Type	Pattern	Hook
Synthesis (combination)	A + B → AB	Marriage
Decomposition	AB → A + B	Divorce
Single replacement	A + BC → AC + B	Someone cuts in on the dance
Double replacement	AB + CD → AD + CB	Partner swap
Combustion	Fuel + O_2 → CO_2 + H_2O	Fire

- **Balancing:** same number of each atom on both sides; change only the coefficients.
- **Catalyst / enzyme** speeds a reaction by lowering activation energy; not used up.

pH

- Scale 0–14. **Below 7 = acid** (more H^+). **7 = neutral** (water). **Above 7 = base** (more OH^-).
- Each step = **10×**. pH 3 is 10× more acidic than pH 4.
- "**A**cids **A**re low; **B**ases are **B**ig numbers."

Solutions: solute (dissolved, e.g. salt) + solvent (does the dissolving, e.g. water) = solution. Water is the "universal solvent."

States of matter

- Solid → liquid = **melting** · liquid → gas = **vaporization** · gas → liquid = **condensation** · liquid → solid = **freezing**.
- Solid → gas = **sublimation** (dry ice). Gas → solid = **deposition** (frost).
- **Triple point** = solid, liquid and gas coexist. **Critical point** = liquid and gas become indistinguishable.

Scientific reasoning (Day 2 · Hour 8)

Scientific method — "Only Quiet Hippos Eat All Cake"

Observe → Question → Hypothesis → Experiment → Analyze → Conclude.

Variables

- **Independent** = what **I** change.
- **Dependent** = what I measure (it **depends** on the independent).
- **Control group** = gets no treatment (for comparison). **Constants** = kept the same.
- Example: "Does caffeine raise heart rate?" → caffeine dose = independent; heart rate = dependent.

Good evidence: large sample, repeated trials, one variable changed at a time. **Correlation ≠ causation.**

Metric — "King Henry Died By Drinking Chocolate Milk"

kilo → hecto → deka → **base** (m, L, g) → deci → centi → milli. Move one step = × or ÷ 10.

- 1 kg = 1,000 g · 1 L = 1,000 mL · 1 m = 100 cm
- Tools: graduated cylinder (volume), balance (mass), thermometer (temp, °C).

Practice quiz (24 questions)

Use 3 questions as each hour's quick quiz, or all 24 timed (24 minutes) in Day 2 · Hour 8. Answers are at the end.

Day 1 · Hour 1 — Foundations

1. Which plane divides the body into left and right? A) Frontal B) Transverse C) Sagittal D) Coronal
2. The wrist is ___ to the elbow. A) Proximal B) Distal C) Medial D) Superior
3. Which is an example of positive feedback? A) Sweating when hot B) Insulin after a meal C) Shivering D) Contractions during labor

Day 1 · Hour 2 — Heart & lungs

4. Blood leaving the right ventricle goes next to the: A) Aorta B) Pulmonary artery C) Left atrium D) Vena cava
5. Which vessel carries oxygenated blood? A) Pulmonary artery B) Vena cava C) Pulmonary vein D) Right ventricle
6. A patient breathing very fast will most likely develop: A) Respiratory acidosis B) Respiratory alkalosis C) Metabolic acidosis D) No change

Day 1 · Hour 3 — Nervous & endocrine

7. Which response is sympathetic? A) Slower heart rate B) Increased digestion C) Dilated pupils D) Constricted bronchioles
8. ADH is made in the ___ and released from the ___. A) Pituitary, hypothalamus B) Hypothalamus, posterior pituitary C) Kidney, adrenal D) Thyroid, parathyroid
9. Which hormone raises blood calcium? A) Calcitonin B) Insulin C) Parathyroid hormone D) Aldosterone

Day 1 · Hour 4 — Digestive & urinary

10. Where does protein digestion begin? A) Mouth B) Stomach C) Small intestine D) Large intestine
11. Bile is made by the ___ and stored by the ___. A) Gallbladder, liver B) Pancreas, liver C) Liver, gallbladder D) Stomach, pancreas
12. Where does blood filtration occur in the kidney? A) Loop of Henle B) Collecting duct C) Glomerulus D) Ureter

Day 2 · Hour 5 — Bones, muscles, skin

13. Which cell breaks down bone? A) Osteoblast B) Osteoclast C) Osteocyte D) Chondrocyte
14. Muscle is attached to bone by: A) Ligaments B) Cartilage C) Tendons D) Fascia
15. Which gland produces sebum? A) Eccrine B) Apocrine C) Sebaceous D) Thyroid

Day 2 · Hour 6 — Immune & reproductive

16. A baby receiving antibodies through breast milk is an example of: A) Natural active B) Artificial active C) Natural passive D) Artificial passive
17. Which cells make antibodies? A) T cells B) B cells C) Neutrophils D) Platelets
18. Fertilization usually happens in the: A) Uterus B) Ovary C) Fallopian tube D) Cervix

Day 2 · Hour 7 — Biology & chemistry

19. Which organelle makes ATP? A) Ribosome B) Golgi C) Mitochondria D) Lysosome
20. Bb × Bb: what percent of offspring show the recessive trait? A) 0% B) 25% C) 50% D) 75%
21. Which particle determines the atomic number? A) Neutron B) Electron C) Proton D) Isotope
22. A solution with pH 3 is: A) Basic B) Neutral C) Acidic D) A salt

Day 2 · Hour 8 — Scientific reasoning

23. Students give plants different amounts of fertilizer and measure height. The dependent variable is: A) Fertilizer amount B) Plant height C) Type of plant D) Sunlight
24. How many milliliters are in 2.5 liters? A) 25 B) 250 C) 2,500 D) 25,000

Answer key

#	Ans	Why
1	C	Sagittal = left/right
2	B	Farther from where the arm attaches
3	D	Labor amplifies until birth
4	B	Right ventricle → pulmonary artery → lungs
5	C	Pulmonary veins bring O_2-rich blood from the lungs
6	B	Fast breathing blows off CO_2 → pH rises
7	C	Fight or flight: big pupils
8	B	Hypothalamus makes, posterior pituitary releases
9	C	PTH pulls calcium up
10	B	Pepsin + acid in the stomach
11	C	Liver makes, gallbladder stores
12	C	Filtering happens at the glomerulus
13	B	Clast = crush
14	C	Tendon = tug
15	C	Sebaceous = sebum
16	C	Borrowed from mom = natural passive
17	B	B cells → antibodies
18	C	Fallopian tube
19	C	Power plant
20	B	Only bb (1 of 4)
21	C	Protons = atomic number
22	C	Below 7 = acid
23	B	Height depends on fertilizer
24	C	1 L = 1,000 mL

Cram sheet: 20 facts to know cold

1. Blood flow: RA → tricuspid → RV → pulmonary artery → lungs → pulmonary vein → LA → mitral → LV → aorta.
2. Pulmonary vein = the only vein with oxygenated blood.
3. Air path: Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveoli.
4. Blood pH 7.35–7.45. Slow breathing → acidosis; fast → alkalosis.
5. Sympathetic = fight or flight; parasympathetic = rest and digest.
6. SAME: Sensory Afferent, Motor Efferent.
7. Cerebellum = balance; medulla = breathing and heart rate.
8. Hypothalamus makes ADH; posterior pituitary releases it; ADH holds water.
9. Calcitonin tones calcium down; PTH pulls it up.
10. Insulin lowers sugar; glucagon raises it.
11. Liver makes bile, gallbladder stores it; bile emulsifies fat.
12. Small intestine absorbs nutrients; large intestine absorbs water.
13. Kidney → ureter → bladder → urethra; filtering at the glomerulus.
14. Blast = build, clast = crush. Ligament = bone–bone; tendon = muscle–bone.
15. Eccrine = cooling sweat; apocrine = odor; sebaceous = oil.
16. Active immunity = your body makes it; passive = borrowed.
17. B cells → antibodies; T cells kill infected cells; IgG crosses placenta.
18. Mitosis = 2 identical (46); meiosis = 4 sex cells (23).
19. A–T, C–G (RNA: A–U). Ionic = give; covalent = share. pH < 7 acid.
20. Independent = what I change; dependent = what I measure.

Test-day tips

- About 1 minute per question. Flag and move on; come back.
- Read “select all that apply” twice — every right answer must be chosen.
- Eliminate two answers first; then choose between the last two.
- No penalty for guessing: never leave a blank.